scientific reports

OPEN

Evaluating the impact of artificial intelligence development on industrial green transformation in China

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Amid the booming development of artificial intelligence (AI) and vigorous green transformation, the influence of AI development on industrial green transformation and its mechanisms requires scientific attention. Utilising a comprehensive set of panel data from 283 prefecture-level cities and above in China from 2004 to 2023, this study methodically examines the direction, mechanisms, heterogeneity characteristics, and spatial effects of AI's impact on industrial green transformation. Results demonstrate that AI development has a significant potential to drive industrial green transformation. This driving effect is primarily manifested through the facilitation of green technological progress and the promotion of economic agglomeration; however, the anticipated channel of energy intensity reduction has not yet been substantiated. Notably, the driving effect of AI exhibits multidimensional heterogeneity, with more pronounced effects in coastal regions, non-resource-based cities, and urban agglomerations. In addition, the investigation reveals that AI development exerts significant positive effects on industrial green transformation within local cities while concurrently exerting negative spatial spillover effects on the industrial green transformation of neighbouring cities. These findings contribute to a more comprehensive theoretical framework and empirical evidence on AI-driven ecological modernisation and offer valuable policy implications for emerging economies seeking to accelerate industrial green transformation through AI technologies.

Keywords Artificial intelligence, Industrial green transformation, Structural green transformation, Efficiency green transformation, Heterogeneity

At present, there is a global consensus on the transition toward green development. The industrial sector, as a core field of energy consumption and pollution emissions, plays a pivotal role in building a modern economic system and achieving sustainable development goals through its green transformation[1,2]. As the world's largest industrial economy, China has made remarkable progress in industrial energy conservation and carbon reduction in recent years. However, it is difficult to fundamentally reverse the long-established extensive development model in the short term. The phase-out and transformation of high-energy-consumption and high-pollution industries still face multiple challenges, rendering the task of industrial green transformation formidable. Meanwhile, artificial intelligence (AI) technologies—represented by smart manufacturing systems, automated control systems, and industrial robots—are flourishing. These technologies have emerged as key drivers in transforming industrial production modes and alleviating resource and environmental constraints. According to data from the Ministry of Industry and Information Technology, China's manufacturing robot density reached 470 units per 10,000 workers in 2023, marking a nearly 19-fold increase over the past decade. For eleven consecutive years, China has remained the world's largest market for industrial robots. The Central Economic Work Conference has repeatedly emphasized the need to "accelerate the development and application of artificial intelligence technologies to support the construction of new industrialization." Against this backdrop, empirically examining whether AI development can effectively enable industrial green transformation is not only a practical necessity for addressing the challenges of China's industrial green transition and integrating technological revolution with industrial transformation, but also an academic imperative for enriching green development theories and providing valuable insights for emerging industrialized nations. This inquiry carries significant theoretical value and practical implications.

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From an academic research perspective, a consensus has been formed regarding the connotation of industrial green transformation. It is widely regarded as a process wherein the industrial sector synergistically evolves toward energy and resource conservation and eco-environmental friendliness, through both intersectoral structural adjustment and improvements in overall development efficiency. This process primarily encompasses two dimensions: structural green transformation and efficiency green transformation[1,3,4]. In terms of measurement approaches, existing studies adopt diverse methodologies: Wang et al.[5] constructed a comprehensive evaluation index system comprising five dimensions, including intensive utilization of energy and resources, pollution reduction, and industrial structure upgrading, and applied the entropy method to achieve comprehensive measurement. Meanwhile, Razzaq et al.[2] adopted GTFP as a metric for industrial green transformation at the provincial level in China, measured by the super-SBM model. These contributions have laid a foundation for subsequent research.

Meanwhile, as a key economic variable, AI has also been extensively explored in terms of its measurement methods and effect studies. At the measurement level, existing studies have employed single indicators such as the number of robots per million hours of work[6], the number of AI patent applications[7], and industrial robot penetration[8,9]. In contrast, other scholars have comprehensively measured the level of AI development from dimensions including infrastructure development, technological application, and market benefits by developing indicator systems[10,11]. In terms of effect studies, scholars have conducted analyses from perspectives including employment structure[12], economic growth[13], global value chains[14], green total factor productivity[15], as well as energy conservation and emissions reduction[16,17]. However, regarding the systematic enabling role of AI development in industrial green transformation, existing literature has not yet provided a clear answer, and there are obvious research gaps: first, there are limitations in research perspectives, as most studies focus on efficiency improvement in a single dimension, resulting in insufficient coverage of the dual connotations of "structure + efficiency" in industrial green transformation; second, the exploration of heterogeneous characteristics is inadequate, and most conclusions are general, failing to provide differentiated guidance for policy-making in different regions; third, research methods are simplistic, with most studies adopting conventional panel data regression and overlooking the potential spatial spillover effects between AI and industrial green transformation; fourth, the sample scale is limited, as most samples are concentrated at the industry, provincial, or national level. The small number of observations easily leads to model instability and overfitting risks, thereby reducing the generalization ability of the research.

Against the above background and in response to the identified research gaps, this study uses panel data from 283 prefecture-level and above cities in China spanning 2004 to 2023 as its research sample. By integrating Python technology, the coupling coordination degree model, the two-way fixed effects model, the two-stage least squares method, and the spatial Durbin model, it systematically examines the direction of impact, mechanism of action, heterogeneous characteristics, and spatial effects of AI development on industrial green transformation. Compared with existing studies, the marginal contributions of this study are as follows: First, it focuses on the dual connotations of "structure + efficiency" in industrial green transformation, conducts an in-depth analysis of the causal relationship and internal mechanism between AI and industrial green transformation, thereby filling gaps in both theoretical and empirical research. Second, it reveals heterogeneous characteristics from three dimensions—geographical location, resource endowment, and spatial organizational forms—providing empirical support for the formulation of differentiated policies. Third, it introduces the spatial Durbin model to capture spatial spillover effects, which better aligns with the spatial relevance of real-world economies and enhances the scientific rigor of the study. Fourth, it acquires city-level raw data through methods such as collecting patent data via Python web scraping and screening enterprise data from Tianyancha (a Chinese enterprise information database), expanding the research scope and increasing the number of sample observations, which significantly improves the reliability and applicability of the empirical results.

Theoretical analysis and research hypotheses

Impact directions of AI development on industrial green transformation

Drawing on existing research, this study defines industrial green transition as a dual-process transition in which both intersectoral structural adjustment and overall development efficiency within the industrial sector shift towards energy conservation, resource efficiency, and environmental sustainability. This transformation primarily involves a structural green transition and an efficiency-driven green transition.

Technological progress plays a key role in this process, with AI, as a frontier of technological progress, having a particularly pronounced impact on driving the structural and efficiency dimensions of green transformation. AI development is fundamentally reshaping industrial production paradigms and steering industrial structures towards greener trajectories. According to Schumpeterian growth theory[18], AI, as a quintessential form of technological innovation, disrupts outdated production modes and sectoral frameworks within industries. It creates new production functions and market demands, expands the frontiers of production possibilities, reduces operating costs, and fosters the emergence of new industrial sectors characterised by green and intelligent features. Based on industrial structure theory, the application of AI technologies accelerates the free flow and reallocation of profit-driven production factors across industrial sectors. Traditional energy-intensive and polluting industries face gradual obsolescence because of their weakened ability to attract high-value production factors while green sectors such as intelligent manufacturing, renewable energy, advanced materials, and information technology experience rapid growth by attracting premium resources, thereby driving structural green transition[19].

Simultaneously, AI enhances production efficiency and facilitates energy conservation and emissions reduction, thereby accelerating efficiency-oriented green transformation. The implementation of AI technologies has significantly increased the levels of automation and intelligence in industrial processes. Through optimised production workflows, intelligent supply chain management, and automated quality inspection, enterprises

achieve resource conservation at fixed output levels, thereby reducing energy consumption and pollutant emissions per unit of output[20]. Resource allocation theory states that scarce resources should be allocated for the most productive uses. AI facilitates this principle through precise data analysis and predictive capabilities, enabling dynamic adjustments to production plans based on market demand forecasts while optimising resource allocation in real time to minimise overproduction and waste. Such optimised allocation not only increases industrial productivity but also significantly reduces the pollution caused by resource misallocation. Accordingly, we propose Hypothesis 1.

Hypothesis 1 AI development contributes to the promotion of industrial green transformation.

Impact mechanisms of AI development on industrial green transformation

Building on existing literature and economic logic, this study posits that the impact mechanisms through which AI development influences industrial green transformation may manifest in the following ways.

First, AI development facilitates green technological progress and thus drives industrial green transformation. AI can catalyse green technology revolutions by accelerating the research, development, and diffusion of clean energy systems, environmentally friendly materials, and pollution control technologies[21]. The advancement of green technologies not only guides traditional industrial sectors towards greener and smarter production paradigms but also fosters the emergence of high-end manufacturing and emerging industries such as renewable energy, advanced materials, green construction, and sustainable transportation. Simultaneously, it optimises industrial production processes, enhances the efficiency of resource use and labour productivity, reduces pollutant emissions per unit of output, and promotes environmentally friendly production models, all of which provide a strong impetus for industrial green transformation.

Second, AI enables the accurate prediction and management of industrial energy demand through intelligent algorithms and big data analytics. These technologies optimise energy allocation and utilisation workflows, improve energy efficiency, achieve cost efficiency, and significantly reduce energy intensity. In line with the imperative requirements of industrial green transformation, lower energy intensity directly reduces energy consumption and pollutant emissions during production[9].

Third, AI development strengthens economic agglomeration effects, thus laying a solid foundation for industrial green transformation. By optimising resource allocation through intelligent algorithms, AI enhances inter-firm collaboration across industrial chains; promotes information sharing and factor mobility among firms; and reduces information asymmetry, communication costs, and transaction costs[22]. These improvements attract greater industrial clustering, which significantly increases agglomeration economies and specialisation. Enhanced agglomeration further amplifies sharing, matching, learning, and spillover effects, which improve energy and resource efficiency, accelerate the diffusion of green technologies and environmental knowledge, and facilitate structural upgrading towards greener industries[23]. In addition, concentrated industrial clusters enable centralised pollution treatment and shared environmental infrastructure, which reduce environmental burdens and promote the transition to greener production modes. Based on these mechanisms, we propose Hypothesis 2.

Hypothesis 2 AI development drives industrial green transformation by mediating green technological progress, reducing energy intensity, and strengthening economic agglomeration.

Heterogeneous effects of AI development on industrial green transformation

Given the significant differences between cities in terms of geographical location, resource endowment, and spatial organisational forms, the heterogeneous impacts of AI development on industrial green transformation under these differentiated contexts requires thorough examination. In terms of geographical location, both AI development and industrial green transformation in China show significant regional differences, with coastal areas objectively outperforming inland regions. Coastal areas benefit from several advantages, including advanced economic development, well-established industrial systems, abundant technological resources, strong policy incentives, and greater environmental awareness. These factors have attracted agglomerations of AI companies and research institutions while leading the nation in initiating industrial relocation and green transformation. Such conditions provide a more favourable external environment for the use of AI to drive industrial green transformation compared with that in inland regions.

In terms of resource endowments, resource-dependent cities often rely heavily on minerals, forestry, and other natural resources for industrial development, with a long-term focus on resource exploitation and utilisation. This path dependency results in homogeneous industrial structures, which pose greater challenges to promoting green industrial transformation[24]. Consequently, the catalytic effect of AI development on industrial green transformation is expected to be weaker in resource-dependent cities than in non-resource-dependent ones.

From the perspective of spatial organisation, urban agglomerations are widely recognised as the most advanced form of regional spatial configuration[25]. Urban agglomerations typically outperform non-agglomeration areas in terms of industrial infrastructure, environmental governance commitment, factor agglomeration, resource accumulation, and market-oriented reforms. In particular, certain urban agglomerations exert significant resource siphoning effects from non-agglomeration areas during AI development and industrial green transformation processes[26], thereby limiting the full realisation of AI's transformative potential in non-agglomeration areas. Based on this analysis, we propose Hypothesis 3.

Hypothesis 3 The positive impact of AI development on industrial green transformation exhibits heterogeneity across geographical locations, resource endowments, and spatial organisational forms.

Spatial effects of AI development on industrial green transformation

In the real economic landscape, no geographical unit exists in isolation; spatial interactions between regions are ubiquitous. Consequently, the impact of AI development on industrial green transformation may extend beyond individual cities with potential spatial spillovers.

On the one hand, AI development could generate positive spatial spillovers. Existing research has shown that AI technologies generate significant demonstration and knowledge spillovers[27]. Cities with advanced AI capabilities can induce neighbouring cities to emulate and adopt AI through labour mobility, technical exchanges, trade partnerships, cross-regional investment, and observational learning. This environment enhances AI development in neighbouring areas and reinforces their catalytic role in industrial green transformation. In addition, close economic and industrial linkages between neighbouring cities enable AI-driven green transitions in one locality to stimulate cross-sectoral collaboration and resource sharing with neighbouring industrial sectors, thereby promoting broader regional sustainability.

On the other hand, AI development can also generate negative spatial externalities. The growth of AI, which relies heavily on highly skilled talent and capital, could trigger resource siphoning effects that deprive neighbouring regions of critical resources, such as skilled labour, finance, and technology, thereby hampering their ability to harness AI for green transformation. Moreover, while advancing local industrial green transformation (especially structural shifts), AI development could facilitate the relocation of polluting and energy-intensive industries to nearby areas, thereby increasing the decarbonisation pressure on neighbouring industrial sectors[28]. Therefore, the following competing research hypotheses are proposed:

Hypothesis 4a AI development exerts positive spatial spillover effects on industrial green transformation.

Hypothesis 4b AI development generates negative spatial spillover effects on industrial green transformation.

Prior to conducting the empirical analyses, as described in the subsequent sections, this study employs a flowchart to delineate the core logical framework of the research (Fig. 1).

Model construction and variable definitions

Model construction

Building on the theoretical analysis, this study establishes a two-way fixed effects model, controlling for city and year fixed effects, as specified in Eq. (1). The objective is to investigate the directional impact of AI development on industrial green transformation through a baseline regression.

$$ln\,igt_{it} = \alpha + \beta ln\,\mathrm{intel}_{it} + \gamma control_{it} + \delta_i + \eta_t + \varepsilon_{it} \tag{1}$$

In Eq. (1), i and t denote city and year indices, respectively; *igt* represents industrial green transformation; *intel* indicates AI development; *control* signifies the vector of control variables; δ_i and η_t correspond to city and

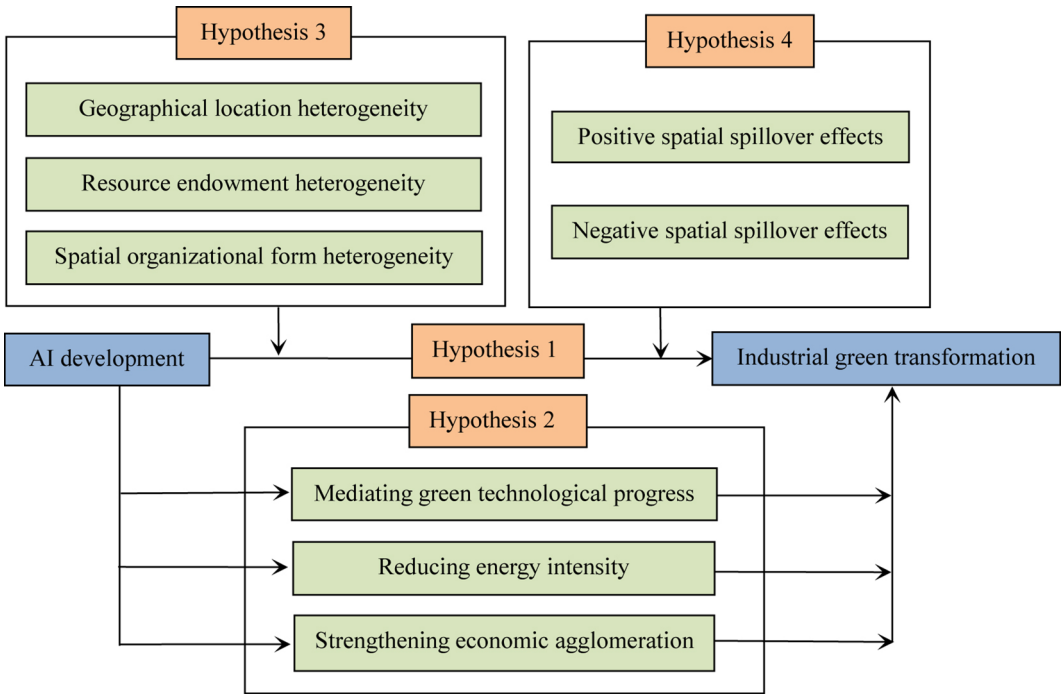

Fig. 1. Logical idea of this study.

time fixed effects, respectively; ε_{it} is the stochastic error term; α denotes the constant term; and β and γ are the parameters to be estimated.

To verify whether AI development facilitates industrial green transformation through mechanisms such as promoting green technological progress, reducing energy intensity, and enhancing economic agglomeration, this study extends Eq. (1) by following the methodology of Chen et al.[29]. Specifically, we regress each mechanism variable M_{it} as the dependent variable, thereby constructing the mechanism test model formalised in Eq. (2).

$$M_{it} = \alpha + \beta ln\,\mathrm{intel}_{it} + \gamma control_{it} + \delta_i + \eta_t + \varepsilon_{it} \tag{2}$$

To examine whether the impact of AI development on industrial green transformation exhibits heterogeneity across geographical locations, resource endowments, and spatial organisational forms, we extend Eq. (1) by categorising the sample cities into subgroups based on these dimensions (as elaborated in subsequent sections). For each dimension of heterogeneity, we designate one reference group and introduce a dummy variable $duma_{it}$ to represent the remaining cities. By incorporating the interaction term $lnintel_{it} \times duma_{it}$ (AI development multiplied by the dummy variable for comparison groups), we construct the heterogeneity test model formalised in Eq. (3).

$$ln\,igt_{it} = \alpha + \beta ln\,\mathrm{intel}_{it} + \lambda ln\,\mathrm{intel}_{it} \times duma_{it} + \gamma control_{it} + \delta_i + \eta_t + \varepsilon_{it} \tag{3}$$

In Eq. (3), λ denotes the parameter to be estimated. To examine whether the impact of AI development on industrial green transformation exhibits spatial spillover effects, we construct the spatial econometric model specified in Eq. (4) for the regression analysis.

$$\begin{aligned} ln\,igt_{it} = \alpha + \rho W \times ln\,igt_{it} + \beta ln\,\mathrm{intel}_{it} + \varphi W \times ln\,\mathrm{intel}_{it} + \gamma control_{it} + \\ \kappa W \times control_{it} + \delta_i + \eta_t + (1 - \tau W)^{-1}\,\varepsilon_{it} \end{aligned} \tag{4}$$

In Eq. (4), W represents an economic–geographical matrix that integrates both economic and geographic factors and is constructed following the methodology of Wang and Guo[30]: Specifically, we first calculate the Euclidean distance based on the longitude and latitude of each city, take its reciprocal as the geographical weight, then use the reciprocal of the absolute difference in average annual per capita regional GDP between cities during the study period as the economic weight, and finally assign a weight of 0.5 to each of these two weight matrices for weighted summation to get the final matrix; ρ is the spatial autocorrelation coefficient; φ and κ are coefficient vectors for the corresponding spatially lagged independent variables; and τ denotes the spatial autocorrelation coefficient of the stochastic error term.

Variable definitions

Industrial green transformation (igt)

Drawing on the general understanding of the connotation of industrial green transformation in academic circles and following the practice in existing literature[1,4,24], this study will comprehensively measure it from two dimensions: structural green transformation and efficiency green transformation. Structural green transition is measured by subtracting from 100% the share of newly registered enterprises in high-polluting industries within the industrial sector. High pollution-load industries are identified based on three criteria that exceed 50% of the total during the sample period: the proportion of industrial SO_2 emissions, chemical oxygen demand (COD) emissions, and general industrial solid waste generation in 20 industry categories and 97 subcategories, as classified by the National Economic Industry Classification (GB/T 4754 – 2017 standard). Ten industries, including electric power and heat generation and supply, are classified as high-polluting industries under this methodology. The efficiency of the green transition is quantified using the reciprocal of SO_2 emissions per unit of industrial value-added. Industrial SO_2 emission is the most widely used indicator in industrial pollution studies to characterise pollution discharge. A reduction in SO_2 emissions per unit of industrial value-added directly reflects improvements in production efficiency and cleaner technological processes within the industrial sector[31]. Given the interdependent relationship between structural and efficiency-driven green transitions, where their coordinated development is essential to achieve holistic industrial green transformation, this study uses the coupling coordination degree model to measure industrial green transformation. This model effectively captures the degree of synergistic correlation between the two dimensions, thereby avoiding both the one-sidedness of single-dimensional measurement and the failure of simple weighted aggregation to reflect the core requirement of "coordinated development." As a result, it produces an assessment of industrial green transformation that more accurately aligns with its composite nature.

AI development (intel)

The current literature on measuring AI development at the city level remains relatively scarce. Existing studies predominantly use metrics such as industrial robot penetration rates or robot density, which are typically estimated using national-level industrial robot stocks in specific sectors combined with city-level employment data for these sectors. However, owing to regional heterogeneity in industrial composition and technology adoption, these estimates may differ significantly from the actual situation in different cities. To address this limitation while balancing data availability and indicator representativeness, this study adopts the approach of Chen and Cai[7] and uses the number of AI patent applications in each city as the primary measure of AI development in the baseline regressions. Although AI patent data may not be an ideal measure given the variations in economic value, technical complexity, commercialisation potential, and diffusion barriers of patents, they remain a robust proxy for large-sample analyses of AI development trajectories. To increase the credibility of the results, we also

use the number of AI firms in each city as an alternative measure of AI development in the robustness checks, thereby mitigating the potential biases inherent in single-indicator approaches.

Mechanism variables (M)
Based on the theoretical framework outlined above, this study selects the following mechanism variables. Green technological progress (*gpat*): Following conventional approaches in the literature, green technological progress is measured by the share of green invention patent applications in the total number of patent applications. Energy intensity (*enin*): Energy intensity is calculated as the ratio of total energy consumption to industrial value-added and captures the energy efficiency of industrial production. Economic agglomeration (*ecoagg*): Economic agglomeration is proxied by enterprise density, defined as the number of above-scale industrial enterprises per unit of built-up area (i.e. the ratio of above-scale industrial enterprises to the area of urban districts).

Control variables (control)
Drawing on established methodologies from previous studies[1,2,4,5], this study incorporates five control variables into the empirical model: (1) industrialisation degree (*indeg*): measured by the ratio of industrial added value to regional GDP; (2) consumption level (*consum*): defined as the ratio of the total retail sales of consumer goods to regional GDP; (3) population density (*peoagg*): calculated as the number of permanent residents per unit of land area; (4) fiscal revenue (*reven*): represented by the ratio of local general budgetary revenue to regional GDP; and (5) openness (*open*): proxied by the proportion of foreign-funded enterprises among above-scale industrial enterprises.

Sample selection and data sources
The analysis uses panel data from 283 cities at the prefecture level and above in China from 2004 to 2023. Cities with significant missing data (e.g. Qinghai, Xinjiang, Tibet, and Hainan) and those in Hong Kong, Macao, and Taiwan are excluded. The data sources are as follows. Newly registered firms and the number of AI firms by industry are retrieved using advanced search filters on Tianyancha (a commercial database). AI patent applications are aggregated to the city level using Python-based techniques to extract and classify data from the China Patent Database following the method of Chen and Cai[7]. The green invention and utility model patents are manually matched using the WIPO IPC Green Inventory from the National Intellectual Property Administration database. The geospatial coordinates for constructing the economic–geographical matrix W are obtained from Baidu Maps. Other variables are compiled from the CEIC, China Economic Net, and EPS databases. Missing values are imputed using linear trend interpolation models, and non-ratio variables are logarithmically transformed. Table 1 presents the descriptive statistics of the main variables.

Analysis of empirical results
Baseline regression analysis
Prior to running the regression analyses, we perform variance inflation factor (*VIF*) tests on the key explanatory variables in the empirical model. The results show that all variables have *VIF* values below 2, with a mean *VIF* of 1.36, which is well below the critical threshold of 10 suggested by empirical guidelines. This value indicates no severe multicollinearity problems. The Hausman test further confirms the appropriateness of using a fixed effects model for the baseline regressions. Consequently, we estimate Eq. (1) using a two-way fixed effects model (controlling for city and year effects) with cluster-robust standard errors to identify the impact directions of AI development on industrial green transformation. The results are summarised in Table 2. Column (1) of Table 2 shows that the coefficient of AI development on industrial green transformation is statistically significant and positive at the 1% level without controlling for covariates. Columns (2) to (6) progressively introduce the control variables, namely, industrialisation degree, consumption level, population density, fiscal revenue, and openness, while maintaining the statistically significant positive coefficient of AI development at the 1% level. These results show that AI development consistently drives green industrial transformation regardless of the inclusion of control variables. Figure 2 provides a visual confirmation of this result. Accordingly, Hypothesis 1 is validated, consistent with Li et al.[32], who identified AI technologies as significant enhancers of green development outcomes.

Variables	Symbol	Mean	Std. dev.	Max	Min
Industrial Green Transformation	*igt*	0.218	0.111	0.982	0.001
AI Development	*intel*	3.461	11.267	311.988	0.004
Green technological progress	*gpat*	2.386	2.065	51.014	0.001
Energy intensity	*enin*	0.025	0.023	0.323	0.002
Economic agglomeration	*ecoagg*	0.406	0.748	15.141	0.001
Industrialization degree	*indeg*	0.400	0.129	1.222	0.076
Consumption level	*consum*	0.363	0.104	0.996	0.041
Population density	*peoagg*	4.319	3.460	34.301	0.047
Fiscal revenue	*reven*	0.069	0.026	0.227	0.018
Openness	*open*	0.043	0.050	0.394	0.001

Table 1. Descriptive statistics of variables.

Variables	ln*igt*					
	(1)	(2)	(3)	(4)	(5)	(6)
ln*intel*	0.076***	0.069***	0.069***	0.067***	0.067***	0.067***
	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)
Indeg		0.602***	0.586***	0.596***	0.594***	0.595***
		(0.080)	(0.084)	(0.083)	(0.083)	(0.083)
Consum			−0.095	−0.088	−0.088	−0.084
			(0.101)	(0.100)	(0.100)	(0.100)
ln*peoagg*				0.138*	0.134*	0.120*
				(0.074)	(0.075)	(0.074)
Reven					−0.772**	−0.785**
					(0.321)	(0.322)
Open						−0.337
						(0.302)
Constant	-2.452***	−2.655***	−2.616***	−3.400***	−3.337***	−3.241***
	(0.081)	(0.088)	(0.095)	(0.422)	(0.425)	(0.417)
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.710	0.717	0.717	0.718	0.718	0.718
Observations	5660	5660	5660	5660	5660	5660

Table 2. Empirical results of baseline model. ***$p < .01$, **$p < .05$, *$p < .1$. The values within the parentheses are clustered robust standard errors.

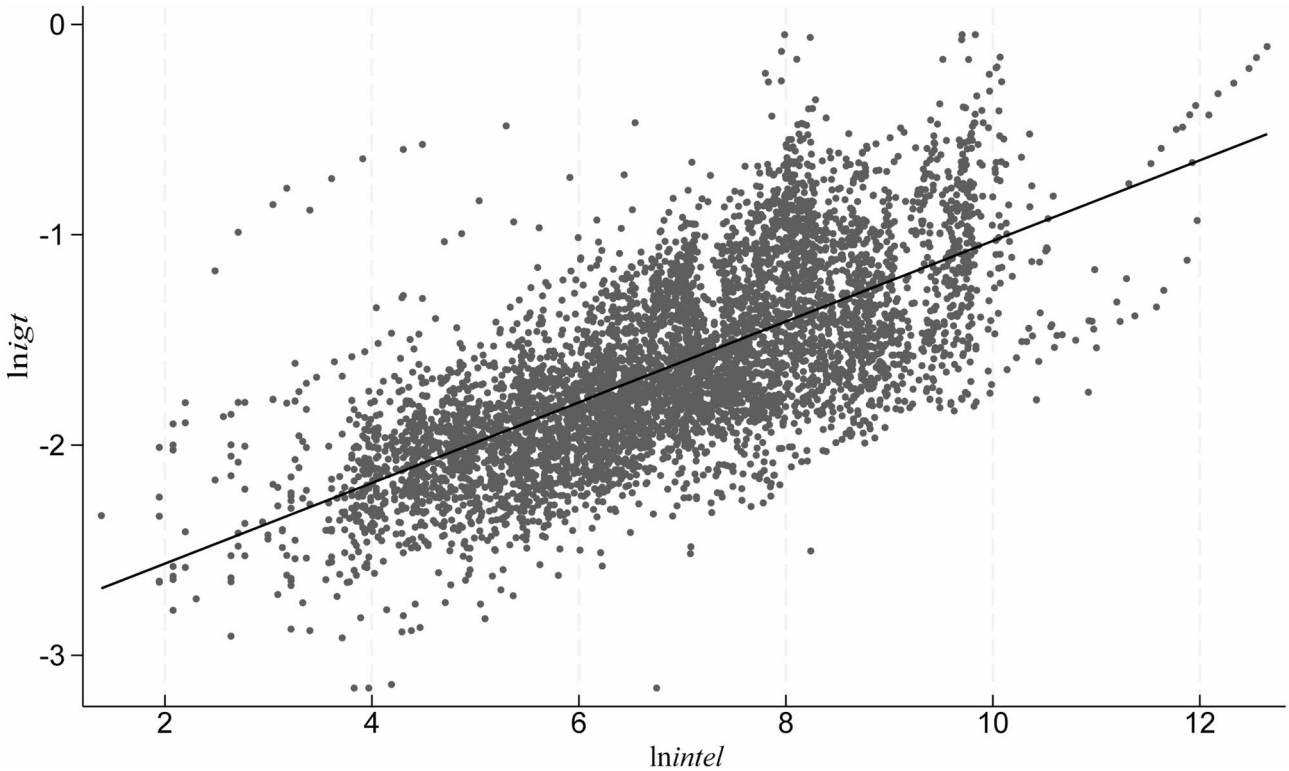

Fig. 2. Scatter points and fitted line of AI development and industrial green transformation.

Regarding the control variables, the most comprehensive empirical results in column (6) of Table 2 show clear patterns: the coefficient on industrialisation degree is statistically significant and positive, suggesting that industrialisation progress significantly promotes industrial green transformation by providing a robust material base for the industrial sector, enabling firms to adopt environmentally friendly technologies and cultivate green corporate identities. Population density has a significantly positive coefficient, suggesting that densely populated

regions accelerate industrial green transformation through increased public environmental sensitivity and social accountability pressures. Fiscal revenue has a significantly negative coefficient, reflecting an inverse relationship with green transition, possibly because of cities' path dependence on traditional industrial models as primary revenue sources, which discourages green transition investments. Finally, the impact coefficients of consumption level and openness are both negative yet not significant, indicating that the effect of improving consumption and openness on industrial green transformation remains uncertain.

Robustness tests

To ensure the robustness of the results, this study conducts a series of sensitivity tests.

First, the dependent variable is substituted. Replacing the composite industrial green transition in the baseline regression with its subdimensions, namely, structural green transition and efficiency-driven green transition, yields the results shown in columns (1) and (2) of Table 3. The coefficients for AI development on both subdimensions remain statistically significant and positive at the 1% level, consistent with the baseline results and confirming the enabling role of AI in industrial green transformation.

Second, alternative core explanatory variables are used. Using the number of AI firms and patent applications for AI inventions (instead of the total number of AI patents) to measure AI development, as shown in columns (3) and (4) of Table 3, yields coefficients that are similarly significant and positive at the 1% level. The results confirm the robustness of the driving effect of AI.

Third, outliers are treated. A 1% winsorisation of the key variables mitigates the influence of extreme values, with the results in column (5) of Table 3 showing no deviation from the baseline coefficient signs.

Fourth, municipalities and provincial capitals are excluded. Removing these cities, which differ significantly from prefecture-level cities in terms of economic foundations, political status, factor agglomeration, and AI application scenarios, yields the results in column (6) of Table 3, where the AI coefficient remains positive and significant at the 1% level.

Fifth, the estimation model is changed. Replacing the two-way fixed effects model with an interactive fixed effects model for Eq. (1), as shown in column (7) of Table 3, improves the control for unobserved factors that vary across cities and time. While the two-way fixed effects model separately accounts for city heterogeneity and time trends, the interactive fixed effects model better accommodates the complexity of the panel data and improves model fit. Additionally, given that the industrial green transformation variable takes non-negative values, this study specifically follow the approach of Lin and Xie[33] to replace the two-way fixed effects with the Tobit model for the parameter estimation of Eq. (1), and the results are shown in Column (8) of Table 3. After testing, the empirical results remain robust.

Endogeneity handling

Although the baseline regression mitigates measurement error endogeneity by prioritising authoritative data sources and addresses omitted variable bias through control variables alongside city and year fixed effects, the potential reverse causality between AI development and industrial green transformation remains unresolved. Specifically, while AI development may drive industrial green transformation, the need for green transformation may also incentivise industrial firms to adopt AI technology in their operations. To address this endogeneity challenge, this study employs two instrumental variables (*IV*s) for AI development and implements a 2SLS approach. First, the average number of AI patent applications in other cities within the same province serves as an *IV*; the results are reported in columns (1) and (2) of Table 4. Second, we use the one-period lagged values of AI development as *IV*s for the current level, as shown in columns (3) and (4). The first-stage regression results show that both *IV*s have significantly positive coefficients at the 1% level, with weak instrument test statistics well above the critical threshold of 10. Meanwhile, the under-identification tests meet the required criteria, thus confirming the validity and relevance of the selected *IV*s. In the second stage, AI development retains a statistically significant positive impact on industrial green transformation after accounting for endogeneity, thereby further confirming the robustness of the baseline findings.

	ist	**ln*iet***	**ln*igt***	**ln*igt***	**ln*igt***	**ln*igt***	**ln*igt***	**ln*igt***
Variables	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**
ln*intel*	3.107***	0.230***	0.069***	0.069***	0.056***	0.072***	0.038**	0.067***
	(0.643)	(0.051)	(0.018)	(0.017)	(0.015)	(0.017)	(0.017)	(0.010)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes		Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes		Yes
City-Year fixed effects							Yes	
R^2	0.395	0.871	0.718	0.719	0.859	0.716	0.183	
Observations	5660	5660	5660	5660	5660	5580	5660	5660

Table 3. Robustness test results. ***$p < .01$, **$p < .05$, *$p < .1$. The values within the parentheses are clustered robust standard errors.

Variables	IV = ln*averintel*		IV = L.ln*intel*	
	First stage	Second stage	First stage	Second stage
	(1) ln*intel*	(2) ln*igt*	(3) ln*intel*	(4) ln*igt*
ln*averintel*	0.671***			
	(0.035)			
L.ln*intel*			0.804***	
			(0.036)	
ln*intel*		0.101***		0.072***
		(0.021)		(0.020)
Control variables	Yes	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Weak instrument test	357.75***		487.90***	
Underidentification Test	109.76***		13.73***	
Observations	5540	5540	5377	5377

Table 4. Results of instrumental variable (*IV*) regression. ***$p < .01$, **$p < .05$, *$p < .1$. The values within the parentheses are clustered robust standard errors.

Variables	*gpat*	ln*enin*	ln*ecoagg*
	(1)	(2)	(3)
ln*intel*	0.456***	−0.022	0.246***
	(0.095)	(0.021)	(0.046)
Control variables	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
R^2	0.153	0.699	0.453
Observations	5660	5660	5660

Table 5. Mechanism test results. ***$p < .01$, **$p < .05$, *$p < .1$. The values within the parentheses are clustered robust standard errors.

Mechanism tests

Table 5 reports the results of the parameter Estimation for Eq. (2) using a two-way fixed effects model with cluster-robust standard errors. Columns (1) to (3) validate the mechanisms through which AI development influences industrial green transformation: green technological progress, energy intensity, and economic agglomeration. As shown in column (1), the coefficient of AI development on green technological progress is statistically significant and positive at the 1% level, indicating that AI facilitates industrial green transformation by promoting green technological progress. Column (2) shows a negative but statistically insignificant coefficient of AI development on energy intensity, suggesting that the potential role of AI in reducing energy intensity remains unrealised, partially weakening its overall contribution to green transformation. A plausible explanation Lies in the fact that the application of AI in the industrial sector is still in its nascent stage. Technology implementation mostly focuses on improving production efficiency rather than optimizing energy consumption, and has not achieved in-depth synergy with the industrial sector's existing energy-saving technologies and production processes. This makes it difficult to translate ai's potential value in reducing energy intensity into practical outcomes, thereby preventing the effective unblocking of transmission channels for promoting industrial green transformation through this pathway. Column (3) shows the significantly positive coefficient of AI development on economic agglomeration at the 1% level, confirming that AI enhances agglomeration economies, which in turn drive industrial green transformation. Collectively, these results provide only partial empirical support for hypothesis 2.

Heterogeneity analysis

Geographical location heterogeneity

To examine how the relationship between AI development and industrial green transformation varies across geographical locations, we divide the sample into coastal cities (113 cities, reference group) and inland cities (170 cities, $duma_{it}$) for parameter estimation according to Eq. (3). The results in column (1) of Table 6 indicate that in line with theoretical expectations, AI development promotes industrial green transformation in both regions, but the effect is stronger in coastal areas (0.078) than in inland areas (0.061, calculated as 0.078-0.017). This difference is due to the advanced economic conditions of coastal regions, dense industrial agglomerations, and unparalleled economies of scale. The concentration of industrial firms in coastal regions allows them to

Variables	Geographical location (1)	Resource endowment (2)	Spatial Organizational Form (3)
ln*intel*	0.078***	0.052***	0.105***
	(0.021)	(0.016)	(0.016)
ln*intel*×*duma*	−0.017***	0.023*	−0.043***
	(0.010)	(0.012)	(0.012)
Control variables	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
R^2	0.719	0.719	0.721
Observations	5660	5660	5660

Table 6. Heterogeneity test results. ***$p < .01$, **$p < .05$, *$p < .1$. The values within the parentheses are clustered robust standard errors.

effectively share the fixed costs associated with AI technology R&D, equipment procurement, and production line upgrades through collaboration and resource sharing, thereby incentivising the green transition through cost efficiency. Additionally, coastal regions have greater agglomeration advantages in terms of capital, talent, and technological resources. Abundant funding supports AI innovation and knowledge diffusion while a robust talent pool facilitates the deeper integration of AI into industrial processes. These factors create a fertile ecosystem for AI-driven green transformation, an advantage that is less pronounced in inland regions.

Resource endowment heterogeneity

To test whether the relationship between AI development and industrial green transformation shows heterogeneity across resource endowments, we classify the sample into resource-dependent cities (114 cities, reference group) and non-resource-dependent cities (169 cities, $duma_{it}$) based on the National Sustainable Development Plan for Resource-Dependent Cities (2013–2020). The parameter estimation according to Eq. (3) yields the results shown in column (2) of Table 6. While AI development significantly drives industrial green transformation in both groups, its effect is weaker in resource-dependent cities (0.052) than in non-resource-dependent ones (0.075, calculated as 0.052˜0.023). This disparity is due to the homogeneous industrial structure and entrenched path dependency of resource-dependent cities where production activities are mainly focused on resource extraction and primary processing. Firms in these cities have limited motivation to adopt AI technologies or pursue structural and efficient green transitions owing to their habitual reliance on energy-intensive practices, thereby limiting the transformative potential of AI.

Spatial organisational form heterogeneity

Urban agglomerations, recognised as the highest form of regional spatial configuration[25], provide a critical context for heterogeneity analyses. We divide the sample into urban agglomeration cities (101 cities, reference group) and non-agglomeration cities (182 cities, $duma_{it}$), with classification based on membership in 10 mature agglomerations: Yangtze River Delta, Pearl River Delta, Beijing–Tianjin–Hebei, Shandong Peninsula, West Coast Economic Zone, Central Plain, Middle Yangtze River, Chengdu–Chongqing, Guanzhong Plain, and Liaodong Peninsula. The results in column (3) of Table 6 show that AI development promotes industrial green transformation in both regions, but the effect is much stronger in urban agglomerations (0.105) than in non-agglomeration areas (0.062, calculated as 0.105-0.043). The advantages of metropolitan areas include rational industrial layouts, complete supply chains, and intercity specialisation, which facilitate the integration of AI across production stages. High population density and expanding consumer markets in these regions drive the demand for green products (e.g. organic food, eco-apparel, and energy-efficient appliances), thus accelerating the replacement of polluting industries with sustainable alternatives. In addition, metropolitan areas concentrate on innovation resources, including research institutions and technology companies, which enable the efficient translation of AI advances into industrial applications that improve production efficiency and energy use and catalyse an efficient green transformation.

In conclusion, the positive impact of AI development on industrial green transformation shows heterogeneous characteristics because of differences in geographical location, resource endowment, and spatial organisational forms. Thus, Hypothesis 3 is confirmed.

Further discussion: analysis of spatial effects

Before conducting the spatial econometric analysis, we perform spatial autocorrelation tests to determine whether the study variables exhibit spatial dependence. Using the global Moran's I under the economic–geographical matrix, we test the spatial autocorrelation of industrial green transformation. The results indicate significant spatial autocorrelation, at least at the 10% confidence level, for all years except 2016, justifying the inclusion of spatial factors. Subsequent model selection tests, including the Hausman, likelihood ratio, and Wald tests, confirm that the SDM with two fixed effects is more appropriate for estimating Eq. (4) than the spatial lag and spatial error models. Table 7 presents the results of the SDM estimation. Column (1) shows a statistically significant negative spatial autocorrelation coefficient (ρ) for industrial green transformation at the 5% level, suggesting negative spatial interdependence. This result may be due to a lack of inter-city cooperation in green

Variables	ln*igt*			
	SDM	Direct	Indirect	Total
	(1)	(2)	(3)	(4)
ln*intel*	0.067***	0.068***	−0.101***	−0.033
	(0.010)	(0.010)	(0.026)	(0.024)
W×ln*intel*	−0.103***			
	(0.027)			
ρ	−0.057**			
	(0.029)			
Control variables	Yes	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
R^2	0.286	0.286	0.286	0.286
Observations	5660	5660	5660	5660

Table 7. SDM Estimation results. ***$p < .01$, **$p < .05$, *$p < .1$. The values within the parentheses are clustered robust standard errors.

transformation efforts, where cities compete for production factors, green firms, and environmental policies, coupled with the tendency of polluting industries to relocate nearby, thus creating negative externalities for neighbouring cities. Column (1) also shows the significantly negative coefficient on the spatially lagged term $W \times \ln intel$ at the 1% level, tentatively indicating negative spatial spillovers from AI development. To disentangle the direct and indirect (spatial spillover) effects, we apply partial differential decomposition[34] to the SDM. The results are presented in columns (2)–(4) of Table 7. The decomposition confirms that AI development significantly promotes green industrial transformation locally but inhibits it in neighbouring cities, thus demonstrating a negative spatial spillover effect. The potential reasons for the negative spatial spillover effect may lie in the fact that AI development in the focal city creates a siphoning effect on core production factors such as high-end talent and technological capital, thereby limiting the resource availability for AI-driven industrial green transformation in surrounding cities. Meanwhile, disparities in digital infrastructure development among cities make it difficult for neighboring areas to benefit from AI technology spillovers. Compounded by the relocation of high-pollution and high-energy-consumption enterprises from the focal city to its peripheries, as well as intensified intercity competition for green policies and high-quality environmentally friendly firms, these factors collectively hinder the industrial green transformation process in adjacent cities. This result supports Hypothesis 4b.

Conclusion and policy implications

Conclusion

This study systematically investigates the directional impacts, mechanisms, heterogeneity, and spatial effects of AI development on industrial green transformation using panel data from 283 Chinese cities at the prefecture level and above (2004–2023). Specifically, it integrates methods such as Python, coupling coordination degree model, two-way fixed effects, interactive fixed effects, 2SLS, and the SDM. The main findings are as follows.

(1) AI development significantly drives green industrial transformation, a conclusion robust to variable substitution, outlier removal, model re-specification, and endogeneity controls.
(2) AI development facilitates green transformation by promoting green technological progress and enhancing economic agglomeration. However, its potential to reduce energy intensity remains unrealised.
(3) Multidimensional heterogeneity exists. Specifically, coastal regions show a stronger AI-driven transformation than inland areas, resource-dependent cities lag behind their non-resource-dependent counterparts, and urban agglomerations outperform non-agglomeration regions.
(4) AI development shows localised promotive effects but generates negative spatial spillovers on the green transformation of neighbouring cities.

Policy implications

The following policy implications emerge based on the research findings.

(1) The demonstrated effectiveness of AI development in driving industrial green transformation provides a theoretical foundation for accelerating AI technology R&D, expanding industrial applications, and formulating innovation-driven policies. Strengthening high-level AI governance frameworks and strategic roadmaps is imperative to ensure seamless integration between AI development strategies and green transition goals. This undertaking requires clearly defined implementation measures, accountable bodies, and precise resource allocation to maximise policy synergies.
(2) Beyond harnessing the direct facilitating effects of AI, policymakers must identify catalytic policies to amplify its mechanism-driven effects, particularly in promoting green technologies, reducing energy intensity, and enhancing agglomeration economies. Priority should be given to the integration of AI and energy sys-

tems to optimise the smart management of energy production, distribution, and consumption and thereby achieve measurable intensity reductions.

(3) Given the multidimensional heterogeneity, AI policies must be context-sensitive and adapted to regional resource endowments, industrial structures, market demands, and urban functional roles. Homogeneous, scale-driven approaches that prioritise short-term gains over sustainable compatibility with local economic realities should be avoided to prevent efficiency losses through misallocated resources.

(4) Proactively mitigating the negative spatial spillovers of AI is critical. Efforts must be exerted to establish intercity coordination mechanisms for AI and green transition initiatives and balance factor allocation across regions to curb excessive resource concentration. Simultaneously, technical and financial support for negatively affected cities should be enhanced, and their AI innovation capacity should be improved to foster endogenous green transition capabilities.

Data availability

The data used in this study will be made available upon request by the corresponding author.

Received: 23 June 2025; Accepted: 6 November 2025
Published online: 22 December 2025

References

1. Meng, X., Xu, S. & Hao, M. Can digital-real integration promote industrial green transformation: fresh evidence from china's industrial sector. *J. Clean. Prod.* **426**, 139116 (2023).
2. Razzaq, A., Sharif, A., Yang, X. & Dogan, E. Influence mechanism of electricity price distortion on industrial green transformation: A Spatial analysis of Chinese regions. *Energy Econ.* **130**, 107308 (2024).
3. Research Group of Institute of Industrial Economics CASS. A study on the green transformation of Chinese industry. *China Industrial Econ.* **4**, 5–14 (2011). (In Chinese).
4. Mao, W., Wang, W. & Sun, H. Driving patterns of industrial green transformation: A multiple regions case learning from China. *Sci. Total Environ.* **697**, 134134 (2019).
5. Wang, E., Cao, Q., Ding, Y. & Sun, H. Fiscal decentralization, government environmental preference and industrial green transformation. *Sustainability* **14**, 14108 (2022).
6. Graetz, G. & Michaels, G. Robots at work. *Rev. Econ. Stat.* **100**, 753–768 (2018).
7. Chen, N. & Cai, Y. AI, accommodating capacity and eonomic Gowth in china: the new Solow paradox and an empirical analysis using the AI patent data. *Econ. Perspect.* **11**, 39–57 (2022). (In Chinese).
8. Acemoglu, D. & Restrepo, P. Robots and jobs: evidence from US labor markets. *J. Polit. Econ.* **128**, 2188–2244 (2020).
9. Yin, Z. H. & Zeng, W. P. The effects of industrial intelligence on china's energy intensity: the role of technology absorptive capacity. *Technol. Forecast. Soc. Chang.* **191**, 122506 (2023).
10. Wang, L. et al. Can industrial intellectualization reduce carbon emissions? — Empirical evidence from the perspective of carbon total factor productivity in China. *Technol. Forecast. Soc. Chang.* **184**, 121969 (2022).
11. Wang, L., Chen, Q., Dong, Z. & Cheng, L. The role of industrial intelligence in peaking carbon emissions in China. *Technol. Forecast. Soc. Chang.* **199**, 123005 (2024).
12. Yang, C. How artificial intelligence technology affects productivity and employment: Firm-level evidence from Taiwan. *Res. Policy*. **51**, 104536 (2022).
13. Wang, Y. & Li, Y. Chinese economic growth and sustainable development: role of artificial intelligence and natural resource management. *Resour. Policy*. **85**, 103996 (2023).
14. Chen, H. & Wang, S. Can the development of industrial intelligence improve the benefits of china's participation in global value chains? *Environ. Impact Assess. Rev.* **105**, 107445 (2024).
15. Xie, Y. & Lin, B. Does AI-driven innovation improve green productivity? The role of heterogeneous information infrastructure. *J. Environ. Manage.* **385**, 126116 (2025).
16. Zhang, W. & Zeng, M. Is artificial intelligence a curse or a blessing for enterprise energy intensity? Evidence from China. *Energy Econ.* **134**, 107561 (2024).
17. Zhang, W., Fan, H. & Zeng, M. Unleashing the power of artificial intelligence: A game changer for urban energy efficiency in China. *Sustainable Cities Soc.* **126**, 106372 (2025).
18. Cozzi, G., Giordani, P. E. & Zamparelli, L. The refoundation of the symmetric equilibrium in schumpeterian growth models. *J. Econ. Theory*. **136**, 788–797 (2007).
19. Xia, L., Han, Q. & Yu, S. Industrial intelligence and industrial structure change: effect and mechanism. *Int. Rev. Economics Finance*. **93**, 1494–1506 (2024).
20. Zhou, Y. & Lin, B. The energy-saving effect of green fiscal policy: empirical evidence from china's comprehensive demonstration cities of energy conservation and emission reduction fiscal policy. *Appl. Energy*. **378**, 124784 (2025).
21. Liu, Y., Shen, F., Guo, J., Hu, G. & Song, Y. Can artificial intelligence technology improve companies' capacity for green innovation? Evidence from listed companies in China. *Energy Econ.* **143**, 108280 (2025).
22. Hanlon, W. W., Miscio, A. & Agglomeration A long-run panel data approach. *J. Urban Econ.* **99**, 1–14 (2017).
23. Liu, J., Qian, Y., Song, S. & Duan, R. Industrial symbiotic agglomeration and green economic growth: A Spatial difference-in-differences approach. *J. Clean. Prod.* **364**, 132560 (2022).
24. Yang, X., Liu, X., Ran, Q. & Razzaq, A. How does natural resource dependence influence industrial green transformation in china? Appraising underlying mechanisms for sustainable development at regional level. *Resour. Policy*. **86**, 104191 (2023).
25. Feng, R., Shen, C., Dai, D. & Xin, Y. Examining the Spatiotemporal evolution, dynamic convergence and drivers of green total factor productivity in china's urban agglomerations. *Economic Anal. Policy*. **78**, 744–764 (2023).
26. Ma, W., Fang, Z. & Zhang, X. Comparative analysis of structural characteristics of china's 18 typical urban agglomerations based on flows of various elements. *Ecol. Model.* **479**, 110308 (2023).
27. Wang, Y., Zhao, W. & Ma, X. The Spatial spillover impact of artificial intelligence on energy efficiency: empirical evidence from 278 Chinese cities. *Energy* **312**, 133497 (2024).
28. Wei, D., Liu, Y. & Zhang, N. Does industry upgrade transfer pollution: evidence from a natural experiment of Guangdong Province in China. *J. Clean. Prod.* **229**, 902–910 (2019).
29. Chen, Y., Fan, Z. Y., Gu, X. M. & Zhou, L. A. Arrival of young talent: the send–down movement and rural education in China. *Am. Econ. Rev.* **110**, 3393–3430 (2020).
30. Wang, J. & Guo, D. Siphon and radiation effects of ICT agglomeration on green total factor productivity: evidence from a Spatial Durbin model. *Energy Econ.* **126**, 106953 (2023).
31. Wang, Y., Ye, H., Wang, Q., Zhou, D. & Su, B. Cleaner production vs end-of-pipe treatment: evidence from industrial SO_2 emissions abatement in China. *J. Environ. Manage.* **277**, 111429 (2021).

32. Li, L., Zhao, J., Yang, Y. & Ma, D. Artificial intelligence and green development well-being: effects and mechanisms in China. *Energy Econ.* **141**, 108094 (2025).
33. Lin, B. & Xie, Y. How feed-in-tariff subsidies affect renewable energy investments in china? New evidence from firm-level data. *Energy* **294**, 130853 (2024).
34. Li, J., Liu, W., Du, L. & Xiao, J. High-tech industry agglomeration and regional green development: an analysis of Spatial Durbin model. *Technol. Forecast. Soc. Chang.* **205**, 123372 (2024).

Acknowledgements
This study was supported by the Hunan Provincial Social Science Achievement Evaluation Committee of China (XSP2023JJC042), the Hunan Provincial Natural Science Foundation of China (2023JJ40266), and the Hunan Provincial Department of Education of China (23B0559). The authors are grateful for the comments from the anonymous reviewers of this paper.

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Funding
This study was supported by the Hunan Provincial Social Science Achievement Evaluation Committee of China (XSP2023JJC042), the Hunan Provincial Natural Science Foundation of China (2023JJ40266), and the Hunan Provincial Department of Education of China (23B0559).

Declarations

Competing interests
The authors declare no competing interests.

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